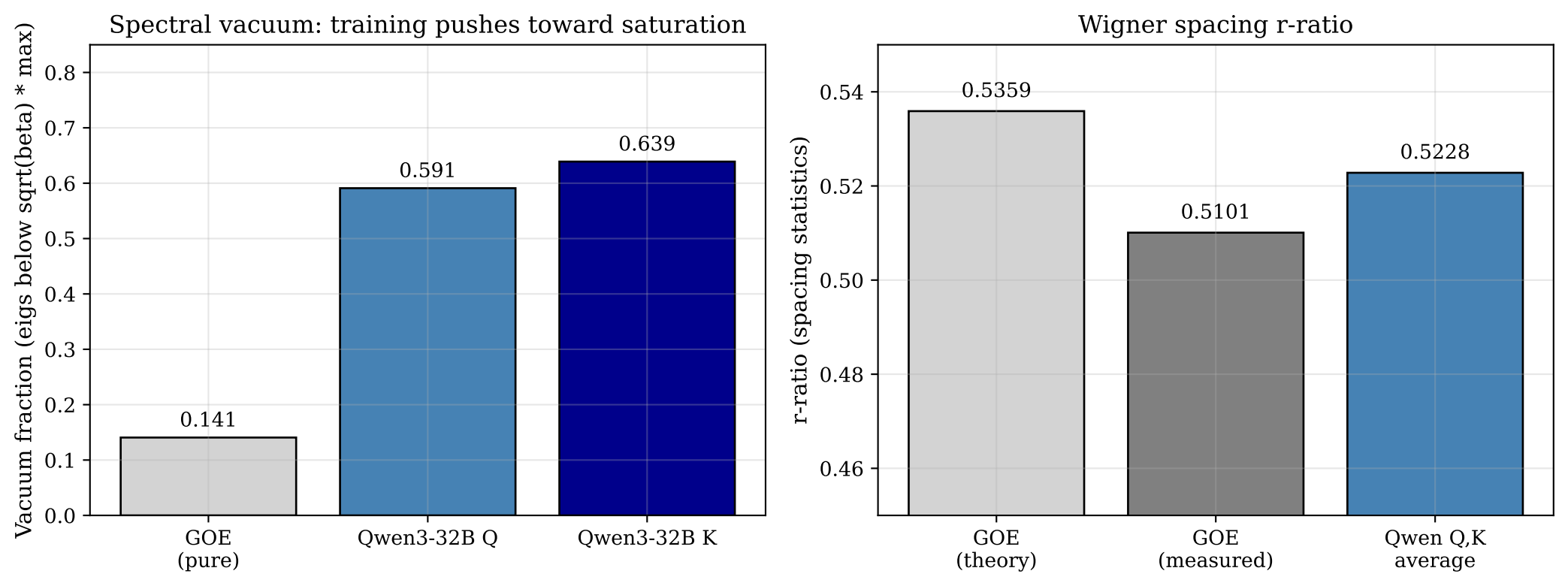
Spectral vacuum: training pushes toward saturation
Vacuum fraction (eigs below sqrt(beta) * max)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.141
0.591
0.639
GOE (pure)
Qwen3-32B Q
Qwen3-32B K
Wigner spacing r-ratio
r-ratio (spacing statistics)
0.46
0.48
0.50
0.52
0.54
0.5359
0.5101
0.5228
GOE (theory)
GOE (measured)
Qwen Q,K average